Yassin Abdulmahdi AI Engineer

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Github | Kaggle

Professional Summary

Data Scientist and Informatics Engineering graduate from Damascus University with hands on experience in AI, machine learning, and data driven problem solving. I specialize in turning challenges into impactful applications

Work Experience

AI Engineer, BeSmart.ai — Feb 2026 – Present | Saudi Arabia

- Building the automation of machine learning lifecycles, from data preprocessing to hyperparameter tuning, to drive the core engine of our AI SaaS platforms.
- Transforming manual AI processes into a fully automated SaaS platform, enabling BeSmart.ai to scale its solutions across multiple industries.

Teaching Assistant, ASPU - Al Sham Private University — Oct 2025 – Present | Damascus

- Teach Machine Learning and Information Retrieval concepts.
- Create tutorial materials and programming examples to support the courses.

Data Scientist, TruePositive — Feb 2025 – Sep 2025 | Dubai (Remote)

- Performed Exploratory Data Analysis (EDA) on subscriber activity and revenue data to uncover churn drivers and usage behavior patterns.
- Delivered Drip, RFM, Growth, and GEO analyses for MTN Congo.
- Worked extensively with the telecom subscriber data, integrating data from subscriber demographics, service engagement snapshots, bundle subscriptions, and revenue facts to create a customer view for advanced analytics.
- Built interactive dashboards, giving client teams real-time visibility into KPIs.
- Social Network Analysis pipeline on large-scale telecom data, reducing processing time by 90% through migration from PySpark to DuckDB.
- Collaborated on projects involving Large Language Models (LLMs).

Research And Development Engineer, — Oct 2024 – Mar 2025
Rachis Systems Sdn. Bhd — Malaysia (Remote)

- Built a federated learning framework with adaptive differential privacy and client clustering.
- Enhanced an iris recognition system by improving accuracy and reliability.
- Applied meta-learning to optimize vehicle routing with time windows.

Education

Bachelor of Science in Information Technology Engineering, — Sep 2019 – Sep 2024
Damascus University

- Relevant Coursework: Advanced Machine Learning, Computer Vision, Natural Language Processing (NLP), Data Structures & Algorithms, and Software Engineering Principles.

Skills

Core AI: Machine Learning, Deep Learning, Computer Vision, NLP, Exploratory Data Analysis

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy

Generative AI & LLMs: RAG, Llama-3, LangChain, LlamaIndex, AutoGen

Tools & Deployment: Docker, Git, Data Storytelling, API Development

Data Engineering & Storage: SQL, LanceDB, DuckDB, Web Scraping

Personal projects

FluentFlow, AI-Powered Public Speaking Coach [Bachelor's Project]

- Engineered a web app that analyzes body language, voice, and speech content, providing AI-driven feedback to improve public speaking skills.
- Integrated Mediapipe + PyTorch for gesture and gaze recognition, Librosa for voice delivery analysis, and NLTK/SpaCy/XGBoost for evaluating script coherence.
- Delivered a Flutter front-end with a Django back-end.

Github ↗ **Demo** ↗

Jigsaw Genius:, Computer Vision-Based Puzzle Solver

Developed an app using advanced computer vision algorithms to solve jigsaw and grid puzzles with high accuracy.

- Solves puzzles in real time with Python + OpenCV.
- Supports hint images to improve accuracy for complex puzzles.
- Interactive, user-friendly interface for easy navigation.
- Technologies: Python, OpenCV.

Github ↗ **Demo** ↗

Intelligent Invoice Line Items Extraction & Retrieval System

end-to-end Extract-Transform-Load and RAG pipeline to automate the extraction and structuring of line-item data from unstructured PDF invoices.

Integrated Docling for high-precision table extraction and LanceDB as a vector store to index serialized row data for semantic search.

Deployed Llama-3 to synthesize grounded answers from retrieved contexts, enabling users to query specific invoice details via natural language.

Github ↗

Sketchy, Interactive Drawing Education System for Children

Developed a web application to improve children's drawing skills by providing real-time, AI-powered feedback. The system uses machine learning to predict and assist in the drawing process, creating an interactive and engaging learning experience.

- Predicts intended drawings and provides guidance.
- Offers completion suggestions to boost creativity and learning.
- Models used: Neural Networks, Deep Learning, KNN, RNN.

Website ↗ **Github** ↗

English Grammar Error Correction

- Fine-tuned a T5 Transformer model within an Encoder-Decoder architecture to detect and correct grammatical errors.
- Designed a simple and accessible interface for users.

Github ↗

Awards

ICPC - International Collegiate Programming Contest

- Ranked 19th in the 2022 Damascus University Collegiate Programming Contest.
- Ranked 12th in the 2021 Al-Baath University Collegiate Programming Contest.
- Achieved **146th place** out of 400+ participants in the 2021 ACPC Kick-off Online Individual Contest.

ICPCOD ↗

Volunteering

Data Scientist, Omdena

AI for Alzheimer's Diagnosis (Omdena Toronto Chapter's project, 2023)

- Developed an ML model to detect early signs of Alzheimer's from brain scan images, focusing on accuracy, efficiency, and deployability.
- Contributed to model optimization, validation, and socially impactful healthcare innovation.

Conversational AI for Transportation Strikes (Omdena Ile-de-France, 2023)

- Built a chatbot to assist citizens with real-time transport alternatives during strikes in France.